

Pico Gym Exclusive - Bit-O-Asm-4

Bit-O-Asm-4

Reverse Engineering Medium 100 pts

by LT 'syreal' Jones

Can you figure out what is in the `eax` register? Put your answer in the picoCTF flag format: `picoCTF{n}` where `n` is the contents of the `eax` register in the decimal number base. If the answer was `0x11` your flag would be `picoCTF{17}`.

In this challenge, we're asked to look at an assembly dump for a program and figure out what's in the `eax` register at the end of the program

Pico Gym Exclusive – Bit-O-Asm-4 x86 Assembly Instructions

But to understand the contents of the disassembly dump file, we'll need to understand a bit about the x86 assembly language



Pico Gym Exclusive – Bit-O-Asm-4 x86 Assembly Instructions

Assembly languages are the programming languages closest to the pure machine code that computer processors execute



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x86 Instructions

```
mov    DWORD PTR [rbp-0x4],0x9fe1a
cmp    DWORD PTR [rbp-0x4],0x2710
jle    0x555555555514e <main+37>
sub    DWORD PTR [rbp-0x4],0x65
jmp    0x5555555555152 <main+41>
```

To solve this challenge, we need to understand three x86 assembly instructions: MOV, CMP, JLE, JMP, and SUB

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The MOV Instruction

```
mov    DWORD PTR [rbp-0x8],0x4
```

MOV <location affected>, <value>

MOV is an assembly instruction which moves data into memory locations

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The MOV Instruction

```
mov     DWORD PTR [rbp-0x8],0x4
```

MOV <location affected>, <value>

In this example, the value 0x4 (decimal 4) is moved into the DWORD PTR [rbp-0x8] location

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The CMP Instruction

```
cmp    DWORD PTR [rbp-0x4],0x2710
```

CMP <location affected>, <value>

The CMP instruction is used to compare a value at a memory location to another value

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The CMP Instruction

```
cmp    DWORD PTR [rbp-0x4],0x2710
```

CMP <location affected>, <value>

After the comparison, another memory location will record whether the LOCATION is greater than, equal to, or less than, the VALUE

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The CMP Instruction

```
cmp    DWORD PTR [rbp-0x4],0x2710
```

CMP <location affected>, <value>

CMP instructions are usually followed by a conditional jump instruction, such as JLE (jump if less than or equal)

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The CMP Instruction

```
cmp    DWORD PTR [rbp-0x4],0x2710
```

CMP <location affected>, <value>

In this example, the contents of the DWORD PTR [rbp-0x4] memory location are compared to the value hexadecimal 2710 (decimal 10000)

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The JLE Instruction

```
jle      0x5555555514e <main+37>
```

JLE <MEMORY ADDRESS>

The JLE instruction jumps to a specified memory address in the program if the previous CMP instruction returned less-than or equal

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The JLE Instruction

```
jle      0x555555555514e <main+37>
```

JLE <MEMORY ADDRESS>

In this example, if the JLE instruction executes, then the program execution will jump to the instruction at memory address 0x5555555555152, which is main function offset +41

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The SUB Instruction

```
sub    DWORD PTR [rbp-0x4],0x65
```

SUB <location affected>, <value>

The SUB instruction subtracts the value of a specified location by a value

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The SUB Instruction

```
sub    DWORD PTR [rbp-0x4],0x65
```

SUB <location affected>, <value>

In this example the value hexadecimal 65 (decimal 101) is subtracted from the value at the memory location DWORD PTR [rbp-0x4]

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The JMP Instruction

```
jmp      0x555555555152 <main+41>
```

JMP <MEMORY ADDRESS>

The JMP instruction jumps to a specified memory address in the program. It is sometimes called an unconditional jump

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The JMP Instruction

```
jmp      0x5555555555152 <main+41>
```

JMP <MEMORY ADDRESS>

In this example, program execution jumps to the instruction at memory address 0x5555555555152, which is offset 41 from the main function

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Keeping Track of Memory Locations

Location	Value
EAX	0
DWORD PTR [rbp-0x4]	0

We will need to track the contents of the memory locations illustrated above

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First Instruction

```
mov     DWORD PTR [rbp-0xc],0x9fe1a  
; rbp-0xc now holds 654874
```

EAX = 0

RBP-0x4 = 654874

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Second Instruction

```
cmp    DWORD PTR [rbp-0x4],0x2710  
; 654874 is compared to 10000
```

EAX = 0

RBP-0x4 = 654874

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Third Instruction

```
jle      0x555555555514e <main+37>  
; this instruction doesn't run  
; since 654874 > 10000
```

EAX = 0

RBP-0x4 = 654874

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Fourth Instruction

```
sub    DWORD PTR [rbp-0x4],0x65  
; 101 is subtracted from 654874
```

EAX = 0

RBP-0x4 = 654773

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Fifth Instruction

```
jmp      0x55555555152 <main+41>  
; program execution jumps to  
; the instruction at <main+41>
```

EAX = 0

RBP-0x4 = 654773

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Sixth Instruction

```
mov     eax,DWORD PTR [rbp-0x4]  
; EAX now holds 654773
```

EAX = 0

RBP-0x4 = 654773